SOURCE CODE

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Student Attendance Tracker</title>

<style>
:root {
  --bg: #f0faf8;
  --teal: #075e5b;
  --teal-light: #d7f4ee;
  --mint: #68d8bd;
  --lime: #d9f75c;
  --coral: #ff8178;
  --white: #ffffff;
  --text: #173b3a;
  --muted: #718987;
  --border: #dcece8;
  --shadow: 0 8px 25px rgba(7, 94, 91, 0.08);
}

* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Arial, sans-serif;
  background: var(--bg);
  color: var(--text);
}

header {
  background: var(--teal);
  color: white;
  padding: 25px 6%;
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
  gap: 15px;
}

header h1 {
  font-size: 27px;
  margin-bottom: 7px;
}

header p {
  color: #c9eee6;
  font-size: 14px;
}

.date-badge {
  background: var(--lime);
  color: var(--teal);
  padding: 12px 18px;
```

```
    border-radius: 14px;
    font-weight: bold;
}

.container {
    max-width: 1250px;
    margin: 30px auto;
    padding: 0 20px;
}

.section-title {
    margin-bottom: 18px;
    font-size: 22px;
}

.stats {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(180px, 1fr));
    gap: 18px;
    margin-bottom: 30px;
}

.stat-card {
    background: var(--white);
    padding: 23px;
    border-radius: 20px;
    box-shadow: var(--shadow);
    border: 1px solid var(--border);
    border-top: 5px solid var(--mint);
}

.stat-card:nth-child(2) {
    border-top-color: #65b7e8;
}

.stat-card:nth-child(3) {
    border-top-color: var(--coral);
}

.stat-card:nth-child(4) {
    border-top-color: var(--lime);
}

.stat-card p {
    color: var(--muted);
    font-size: 14px;
    margin-bottom: 12px;
}

.stat-card h2 {
    font-size: 32px;
    color: var(--teal);
}

.panel {
    background: var(--white);
    border-radius: 20px;
    padding: 25px;
    margin-bottom: 25px;
```

```
  box-shadow: var(--shadow);
  border: 1px solid var(--border);
}

.form-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(190px, 1fr));
  gap: 15px;
}

label {
  display: block;
  font-size: 13px;
  font-weight: bold;
  margin-bottom: 8px;
}

input, select {
  width: 100%;
  padding: 12px;
  border: 1px solid var(--border);
  border-radius: 10px;
  background: #fbfffe;
  color: var(--text);
  outline: none;
  font-size: 14px;
}

input:focus, select:focus {
  border-color: var(--mint);
  box-shadow: 0 0 0 3px #68d8bd30;
}

button {
  border: none;
  padding: 11px 16px;
  border-radius: 10px;
  font-weight: bold;
  cursor: pointer;
  transition: 0.2s;
}

button:hover {
  transform: translateY(-1px);
  opacity: 0.9;
}

.primary-btn {
  background: var(--teal);
  color: white;
}

.mint-btn {
  background: var(--mint);
  color: var(--teal);
}

.coral-btn {
  background: #ffe2df;
```

```
    color: #a83d36;
}

.lime-btn {
    background: var(--lime);
    color: var(--teal);
}

.form-actions {
    display: flex;
    gap: 10px;
    margin-top: 18px;
    flex-wrap: wrap;
}

.toolbar {
    display: flex;
    gap: 12px;
    margin-bottom: 18px;
    flex-wrap: wrap;
}

.toolbar input {
    flex: 1;
    min-width: 180px;
}

.toolbar select {
    width: auto;
    min-width: 150px;
}

.table-wrap {
    width: 100%;
    overflow-x: auto;
}

table {
    width: 100%;
    border-collapse: collapse;
    min-width: 650px;
}

th, td {
    padding: 14px 12px;
    text-align: left;
    border-bottom: 1px solid var(--border);
    font-size: 14px;
}

th {
    background: var(--teal-light);
    color: var(--teal);
}

tbody tr:hover {
    background: #f5fbf9;
}

.badge {
```

```
    display: inline-block;
    padding: 6px 10px;
    border-radius: 20px;
    font-size: 12px;
    font-weight: bold;
}

.Present {
    background: #d7f8e7;
    color: #167547;
}

.Absent {
    background: #ffe0df;
    color: #b33434;
}

.Late {
    background: #fff0c4;
    color: #896000;
}

.action-buttons {
    display: flex;
    gap: 7px;
}

.action-buttons button {
    padding: 7px 10px;
    font-size: 12px;
}

.alert {
    padding: 14px 16px;
    background: #fff0ed;
    border-left: 5px solid var(--coral);
    border-radius: 10px;
    margin-bottom: 20px;
    color: #8c3934;
    line-height: 1.6;
}

.success-message {
    display: none;
    padding: 12px;
    margin-top: 15px;
    border-radius: 10px;
    background: #d7f8e7;
    color: #167547;
}

.empty {
    text-align: center;
    color: var(--muted);
    padding: 25px;
}

footer {
    text-align: center;
    padding: 25px;
```

```
    color: var(--muted);
    font-size: 13px;
}

@media (max-width: 600px) {
    header {
        padding: 22px;
    }

    header h1 {
        font-size: 22px;
    }

    .container {
        padding: 0 12px;
        margin-top: 20px;
    }

    .panel {
        padding: 17px;
    }

    .toolbar select {
        width: 100%;
    }
}
</style>
</head>

<body>

<header>
    <div>
        <h1>🎓 Student Attendance Tracker</h1>
        <p>Smart attendance management for your
classroom</p>
    </div>
    <div class="date-badge" id="todayDate"></div>
</header>

<main class="container">

    <h2 class="section-title">📊 Attendance Dashboard</h2>

    <section class="stats">
        <div class="stat-card">
            <p>Total Students</p>
            <h2 id="totalStudents">0</h2>
        </div>

        <div class="stat-card">
            <p>Present Today</p>
            <h2 id="presentCount">0</h2>
        </div>

        <div class="stat-card">
            <p>Absent Today</p>
            <h2 id="absentCount">0</h2>
        </div>
```

```
  <div class="stat-card">
    <p>Late Today</p>
    <h2 id="lateCount">0</h2>
  </div>
</section>

<section class="panel">
  <h2 class="section-title" id="formTitle">➕ Add New Student</h2>

  <form id="studentForm">
    <div class="form-grid">
      <div>
        <label for="studentId">Student ID</label>
        <input id="studentId" type="text"
            placeholder="e.g. STU101" required>
      </div>

      <div>
        <label for="studentName">Student Name</label>
        <input id="studentName" type="text"
            placeholder="Enter full name" required>
      </div>

      <div>
        <label for="studentClass">Class / Department</label>
        <input id="studentClass" type="text"
            placeholder="e.g. B.Sc Computer Science" required>
      </div>

      <div>
        <label for="studentEmail">Email (Optional)</label>
        <input id="studentEmail" type="email"
            placeholder="student@example.com">
      </div>
    </div>

    <div class="form-actions">
      <button class="primary-btn" type="submit" id="saveBtn">
        Add Student
      </button>
      <button class="coral-btn" type="button" id="cancelEditBtn"
          style="display:none">
        Cancel Edit
      </button>
    </div>

    <div class="success-message" id="message"></div>
  </form>
</section>

<section class="panel">
  <h2 class="section-title">👩‍🎓 Student Directory</h2>

  <div class="toolbar">
    <input id="searchInput" type="search"
```

```
          placeholder="🔍 Search by ID, name or class...">

        <select id="classFilter">
          <option value="all">All Classes</option>
        </select>
      </div>

      <div class="table-wrap">
        <table>
          <thead>
            <tr>
              <th>Student ID</th>
              <th>Name</th>
              <th>Class</th>
              <th>Email</th>
              <th>Actions</th>
            </tr>
          </thead>
          <tbody id="studentTable"></tbody>
        </table>
      </div>
    </section>

    <section class="panel">
      <h2 class="section-title">📝 Mark Daily Attendance</h2>

      <div class="toolbar">
        <div>
          <label for="attendanceDate">Attendance Date</label>
          <input id="attendanceDate" type="date">
        </div>

        <div>
          <label for="attendanceFilter">Show Students</label>
          <select id="attendanceFilter">
            <option value="all">All Students</option>
            <option value="Present">Present</option>
            <option value="Absent">Absent</option>
            <option value="Late">Late</option>
            <option value="Unmarked">Not Marked</option>
          </select>
        </div>
      </div>

      <div class="table-wrap">
        <table>
          <thead>
            <tr>
              <th>Student ID</th>
              <th>Name</th>
              <th>Class</th>
              <th>Attendance Status</th>
              <th>Mark Attendance</th>
            </tr>
          </thead>
          <tbody id="attendanceTable"></tbody>
        </table>
      </div>
```

```
      <div class="form-actions">
        <button class="mint-btn" id="markAllPresentBtn" type="button">
          ✓ Mark All Present
        </button>

        <button class="lime-btn" id="saveAttendanceBtn" type="button">
          Save Attendance
        </button>
      </div>

      <div class="success-message" id="attendanceMessage"></div>
    </section>

    <section class="panel">
      <h2 class="section-title">⚠️ Low Attendance Alerts</h2>
      <p style="color:var(--muted); margin-bottom:15px; line-height:1.6">
        Students with attendance below 75% will appear here.
      </p>
      <div id="lowAttendanceList"></div>
    </section>

    <section class="panel">
      <h2 class="section-title">📅 Attendance History</h2>

      <div class="toolbar">
        <select id="historyStudent">
          <option value="all">All Students</option>
        </select>

        <button class="primary-btn" id="exportBtn" type="button">
          ⬇ Export CSV Report
        </button>
      </div>

      <div class="table-wrap">
        <table>
          <thead>
            <tr>
              <th>Date</th>
              <th>Student ID</th>
              <th>Name</th>
              <th>Class</th>
              <th>Status</th>
            </tr>
          </thead>
          <tbody id="historyTable"></tbody>
        </table>
      </div>
    </section>

  </main>

  <footer>
    Student Attendance Tracker © 2026 | Simple • Smart • Organized
```

```
</footer>

<script>
const $ = id => document.getElementById(id);

let students = [];
let attendance = {};
let editId = null;

/* LOAD SAVED DATA */

try {
  students = JSON.parse(localStorage.getItem("attendance_students")) || [];
  attendance = JSON.parse(localStorage.getItem("attendance_records")) || {};
} catch (error) {
  students = [];
  attendance = {};
}

/* DATE SETUP */

function getToday() {
  const now = new Date();
  const offset = now.getTimezoneOffset();
  return new Date(now.getTime() - offset * 60000)
    .toISOString().slice(0, 10);
}

$("todayDate").textContent = new Date().toLocaleDateString("en-IN", {
  day: "2-digit",
  month: "short",
  year: "numeric"
});

$("attendanceDate").value = getToday();

/* STORAGE */

function saveData() {
  localStorage.setItem("attendance_students", JSON.stringify(students));
  localStorage.setItem("attendance_records", JSON.stringify(attendance));
}

/* MESSAGE */

function showMessage(elementId, message) {
  const box = $(elementId);
  box.textContent = message;
  box.style.display = "block";

  setTimeout(() => {
    box.style.display = "none";
  }, 3000);
}
```

```
/* ATTENDANCE HELPERS */

function getRecord(date, id) {
  return attendance[date] && attendance[date][id]
    ? attendance[date][id]
    : "";
}

function setRecord(date, id, status) {
  if (!attendance[date]) {
    attendance[date] = {};
  }

  attendance[date][id] = status;
}

/* DASHBOARD STATISTICS */

function renderStats() {
  const date = $("attendanceDate").value || getToday();
  const records = attendance[date] || {};

  $("totalStudents").textContent = students.length;

  $("presentCount").textContent = students.filter(
    student => records[student.id] === "Present"
  ).length;

  $("absentCount").textContent = students.filter(
    student => records[student.id] === "Absent"
  ).length;

  $("lateCount").textContent = students.filter(
    student => records[student.id] === "Late"
  ).length;
}

/* STUDENT DIRECTORY */

function renderStudents() {
  const search = $("searchInput").value.toLowerCase();
  const classValue = $("classFilter").value;

  const filtered = students.filter(student => {
    const matchesSearch =
      student.id.toLowerCase().includes(search) ||
      student.name.toLowerCase().includes(search) ||
      student.className.toLowerCase().includes(search) ||
      student.email.toLowerCase().includes(search);

    const matchesClass =
      classValue === "all" || student.className === classValue;

    return matchesSearch && matchesClass;
  });

  const tbody = $("studentTable");
```

```
  if (filtered.length === 0) {
    tbody.innerHTML =
      '<tr><td colspan="5" class="empty">No students found. Add a student to get started!</td></tr>';
  } else {
    tbody.innerHTML = filtered.map(student => `
      <tr>
        <td><strong>${escapeHTML(student.id)}</strong></td>
        <td>${escapeHTML(student.name)}</td>
        <td>${escapeHTML(student.className)}</td>
        <td>${escapeHTML(student.email || "-")}</td>
        <td>
          <div class="action-buttons">
            <button class="mint-btn"
              onclick="editStudent('${encodeURIComponent(student.id)}')">
              Edit
            </button>
            <button class="coral-btn"
              onclick="deleteStudent('${encodeURIComponent(student.id)}')">
              Delete
            </button>
          </div>
        </td>
      </tr>
    `).join("");
  }

  renderClassFilter();
  renderHistoryOptions();
}

/* CLASS FILTER */

function renderClassFilter() {
  const select = $("classFilter");
  const current = select.value;

  const classes = [...new Set(students.map(s => s.className))]
    .sort((a, b) => a.localeCompare(b));

  select.innerHTML = '<option value="all">All Classes</option>';

  classes.forEach(className => {
    const option = document.createElement("option");
    option.value = className;
    option.textContent = className;
    select.appendChild(option);
  });

  if (classes.includes(current)) {
    select.value = current;
  }
}

/* ADD / UPDATE STUDENT */
```

```
$("studentForm").addEventListener("submit", function(event) {
  event.preventDefault();

  const id = $("studentId").value.trim();
  const name = $("studentName").value.trim();
  const className = $("studentClass").value.trim();
  const email = $("studentEmail").value.trim();

  if (!id || !name || !className) {
    showMessage("message", "Please fill in all required fields.");
    return;
  }

  const duplicate = students.some(student =>
    student.id.toLowerCase() === id.toLowerCase() &&
    student.id !== editId
  );

  if (duplicate) {
    showMessage("message", "This Student ID already exists.");
    return;
  }

  if (editId !== null) {
    const index = students.findIndex(student => student.id === editId);

    if (index !== -1) {
      const oldId = students[index].id;

      students[index] = { id, name, className, email };

      if (oldId !== id) {
        Object.keys(attendance).forEach(date => {
          if (attendance[date][oldId] !== undefined) {
            attendance[date][id] = attendance[date][oldId];
            delete attendance[date][oldId];
          }
        });
      }
    }

    showMessage("message", "Student details updated successfully!");
  } else {
    students.push({ id, name, className, email });
    showMessage("message", "Student added successfully!");
  }

  students.sort((a, b) => a.name.localeCompare(b.name));

  saveData();
  resetForm();
  renderAll();
});
```

```
/* EDIT STUDENT */

function editStudent(encodedId) {
  const id = decodeURIComponent(encodedId);
  const student = students.find(item => item.id === id);

  if (!student) return;

  editId = id;

  $("studentId").value = student.id;
  $("studentName").value = student.name;
  $("studentClass").value = student.className;
  $("studentEmail").value = student.email;

  $("formTitle").textContent = "✏️ Edit Student Details";
  $("saveBtn").textContent = "Update Student";
  $("cancelEditBtn").style.display = "inline-block";

  $("studentForm").scrollIntoView({
    behavior: "smooth",
    block: "start"
  });
}

/* DELETE STUDENT */

function deleteStudent(encodedId) {
  const id = decodeURIComponent(encodedId);
  const student = students.find(item => item.id === id);

  if (!student) return;

  if (!confirm("Delete " + student.name + "? This cannot be undone.")) {
    return;
  }

  students = students.filter(item => item.id !== id);

  Object.keys(attendance).forEach(date => {
    if (attendance[date]) {
      delete attendance[date][id];
    }
  });

  saveData();
  renderAll();
}

/* RESET FORM */

function resetForm() {
  $("studentForm").reset();
  editId = null;

  $("formTitle").textContent = "➕ Add New Student";
  $("saveBtn").textContent = "Add Student";
  $("cancelEditBtn").style.display = "none";
}
```

```
$("cancelEditBtn").addEventListener("click", resetForm);

/* ATTENDANCE TABLE */

function renderAttendance() {
  const date = $("attendanceDate").value || getToday();
  const filter = $("attendanceFilter").value;

  const filtered = students.filter(student => {
    const status = getRecord(date, student.id);

    if (filter === "all") return true;
    if (filter === "Unmarked") return !status;

    return status === filter;
  });

  const tbody = $("attendanceTable");

  if (filtered.length === 0) {
    tbody.innerHTML =
      '<tr><td colspan="5" class="empty">No students match this filter.</td></tr>';
    return;
  }

  tbody.innerHTML = filtered.map(student => {
    const status = getRecord(date, student.id);

    return `
      <tr>
        <td><strong>${escapeHTML(student.id)}</strong></td>
        <td>${escapeHTML(student.name)}</td>
        <td>${escapeHTML(student.className)}</td>
        <td>
          <span class="badge ${status || ""}">
            ${status || "Not Marked"}
          </span>
        </td>
        <td>
          <select
            aria-label="Attendance for ${escapeHTML(student.name)}"
            onchange="changeAttendance('${encodeURIComponent(student.id)}', this.value)">
            <option value="" ${status === "" ? "selected" : ""}>
              Not Marked
            </option>
            <option value="Present" ${status === "Present" ? "selected" : ""}>
              Present
            </option>
            <option value="Absent" ${status === "Absent" ? "selected" : ""}>
              Absent
            </option>
            <option value="Late" ${status === "Late" ? "selected" : ""}>
```

```
                    Late
                  </option>
                </select>
              </td>
            </tr>
          `;
    }).join("");
}

/* CHANGE ATTENDANCE */

function changeAttendance(encodedId, status) {
    const id = decodeURIComponent(encodedId);
    const date = $("attendanceDate").value;

    if (!date) {
        showMessage("attendanceMessage", "Please select a date first.");
        return;
    }

    if (status) {
        setRecord(date, id, status);
    } else if (attendance[date]) {
        delete attendance[date][id];
    }

    saveData();
    renderAll();
}

/* MARK ALL PRESENT */

$("markAllPresentBtn").addEventListener("click", function() {
    const date = $("attendanceDate").value;

    if (!date) {
        showMessage("attendanceMessage", "Please select a date first.");
        return;
    }

    if (students.length === 0) {
        showMessage("attendanceMessage", "Add students before marking attendance.");
        return;
    }

    students.forEach(student => setRecord(date, student.id, "Present"));

    saveData();
    renderAll();

    showMessage("attendanceMessage", "All students marked present!");
});

/* SAVE ATTENDANCE */
```

```
$("saveAttendanceBtn").addEventListener("click", function() {
  saveData();
  renderAll();
  showMessage("attendanceMessage", "Attendance saved successfully!");
});

/* LOW ATTENDANCE ALERT */

function renderLowAttendance() {
  const list = $("lowAttendanceList");

  const totalDays = Object.keys(attendance).filter(date =>
    Object.keys(attendance[date] || {}).length > 0
  ).length;

  if (totalDays === 0) {
    list.innerHTML =
      '<p class="empty">Attendance records will appear here after you mark attendance.</p>';
    return;
  }

  const lowStudents = students.map(student => {
    let presentDays = 0;
    let markedDays = 0;

    Object.keys(attendance).forEach(date => {
      const status = getRecord(date, student.id);

      if (status) {
        markedDays++;

        if (status === "Present" || status === "Late") {
          presentDays++;
        }
      }
    });

    const percentage = markedDays
      ? (presentDays / markedDays) * 100
      : 0;

    return { ...student, percentage, markedDays };
  }).filter(student =>
    student.markedDays > 0 && student.percentage < 75
  );

  if (lowStudents.length === 0) {
    list.innerHTML =
      '<p class="empty">🎉 No low-attendance alerts! All recorded attendance percentages are 75% or above.</p>';
    return;
  }

  list.innerHTML = lowStudents.map(student => `
    <div class="alert">
      <strong>${escapeHTML(student.name)}</strong>
      (${escapeHTML(student.id)}) — Attendance:
      <strong>${student.percentage.toFixed(1)}%</strong>
```

```
        <br>
        <small>${student.markedDays} recorded attendance day(s)</small>
      </div>
    `).join("");
}

/* HISTORY DROPDOWN */

function renderHistoryOptions() {
  const select = $("historyStudent");
  const current = select.value;

  select.innerHTML = '<option value="all">All Students</option>';

  students.forEach(student => {
    const option = document.createElement("option");
    option.value = student.id;
    option.textContent = student.name + " (" + student.id + ")";
    select.appendChild(option);
  });

  if (students.some(student => student.id === current)) {
    select.value = current;
  }
}

/* ATTENDANCE HISTORY */

function renderHistory() {
  const selectedId = $("historyStudent").value;
  const rows = [];

  Object.keys(attendance).sort().reverse().forEach(date => {
    students.forEach(student => {
      const status = getRecord(date, student.id);

      if (!status) return;

      if (selectedId !== "all" && selectedId !== student.id) {
        return;
      }

      rows.push({
        date,
        student,
        status
      });
    });
  });

  const tbody = $("historyTable");

  if (rows.length === 0) {
    tbody.innerHTML =
      '<tr><td colspan="5" class="empty">No attendance history available yet.</td></tr>';
    return;
```

```
}

  tbody.innerHTML = rows.map(row => `
    <tr>
      <td>${escapeHTML(row.date)}</td>
      <td>${escapeHTML(row.student.id)}</td>
      <td>${escapeHTML(row.student.name)}</td>
      <td>${escapeHTML(row.student.className)}</td>
      <td>
        <span class="badge ${row.status}">
          ${row.status}
        </span>
      </td>
    </tr>
  `).join("");
}

/* CSV REPORT EXPORT */

$("exportBtn").addEventListener("click", function() {
  const rows = [
    ["Date", "Student ID", "Student Name", "Class", "Status"]
  ];

  Object.keys(attendance).sort().forEach(date => {
    students.forEach(student => {
      const status = getRecord(date, student.id);

      if (status) {
        rows.push([
          date,
          student.id,
          student.name,
          student.className,
          status
        ]);
      }
    });
  });

  if (rows.length === 1) {
    alert("No attendance records available to export.");
    return;
  }

  const csv = rows.map(row =>
    row.map(value =>
      '"' + String(value).replace(/"/g, '""') + '"'
    ).join(",")
  ).join("\r\n");

  const blob = new Blob(["\uFEFF" + csv], {
    type: "text/csv;charset=utf-8;"
  });

  const url = URL.createObjectURL(blob);
  const link = document.createElement("a");

  link.href = url;
  link.download = "student_attendance_report.csv";
```

```
    document.body.appendChild(link);
    link.click();
    link.remove();

    URL.revokeObjectURL(url);
});

/* HTML ESCAPING */

function escapeHTML(value) {
    return String(value).replace(/[&<>"']/g, char => ({
        "&": "&amp;",
        "<": "&lt;",
        ">": "&gt;",
        '"': "&quot;",
        "'": "&#39;"
    })[char]);
}

/* SEARCH AND FILTER EVENTS */

$("searchInput").addEventListener("input", renderStudents);
$("classFilter").addEventListener("change", renderStudents);

$("attendanceDate").addEventListener("change", function() {
    renderStats();
    renderAttendance();
});

$("attendanceFilter").addEventListener("change", renderAttendance);
$("historyStudent").addEventListener("change", renderHistory);

/* RENDER ALL SECTIONS */

function renderAll() {
    renderStats();
    renderStudents();
    renderAttendance();
    renderLowAttendance();
    renderHistory();
}

/* INITIALIZE APPLICATION */

renderAll();
</script>

</body>
</html>
```